Phase 1 Plan: StudyBot Study

Does low-effort AI help cause over-reliance, and can an adaptive tutor prevent it?

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For Phase 1 (due October 25) we submit a working AI-tutor app and the plan for our experiment. Each participant does one 30-minute sitting, and we collect data from October 26 to 28. The study extends Srinivasaprasad et al. (2026), *How AI Chatbot Response Style Affects Cognitive Load and Performance in Educational Tasks*, which found that concise, structured AI answers give the lowest mental effort and the highest accuracy, but only tested scripted answers in a single exposure.

1 At a glance

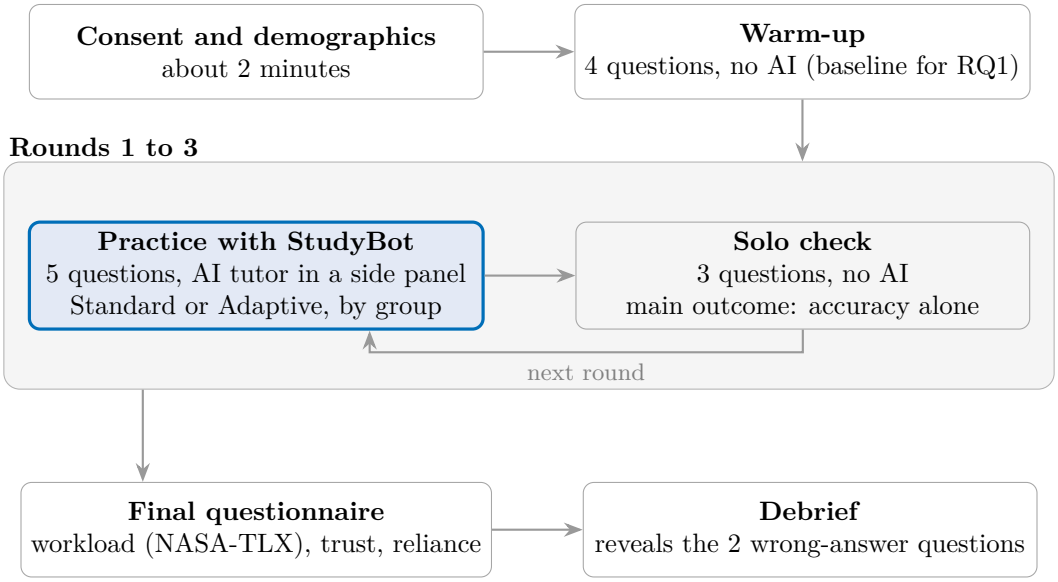


Figure 1: What one participant goes through: 28 questions in about 30 minutes. A one-question effort rating follows every part.

The two groups differ only in the highlighted practice step. Everything else is identical, so any difference in solo-check accuracy comes from the tutor.

2 What we built

StudyBot is a web app that runs the whole experiment. It shows the questions, has a chat panel with a real AI tutor, and records every answer, chat message and rating automatically. The original paper used scripted answers typed in by researchers (Wizard-of-Oz); ours are generated live.

Participants are split evenly into two groups. Both tutors look the same: short bullet points with emojis, about 100 words, the style the paper found easiest on the mind. They differ only in how much they give away.

	Standard tutor	Adaptive tutor (our proposed fix)
Before asking	Chat is open straight away	Student must pick a guess first (or tap "I have no idea yet")
First reply	Always the answer plus a short reason	Depends on how well the student has been doing alone: the answer, a hint, or a guiding question
Follow-up messages	More explanation	Each one gives more help, so the answer is at most 3 messages away

Table 1: How the two tutor versions differ.

Adaptive rule. The adaptive tutor looks at the student's last 4 answers given without AI help. Below 50% correct it gives the full answer; at 50–74% it gives a hint that rules out wrong options; at 75% or more it asks a guiding question.

Wrong-answer checks. On 2 of the 15 practice questions, whenever the tutor gives a full answer it is a plausible but wrong one. This shows whether students check the AI or simply accept it. The debrief at the end tells them which questions these were.

Questions. We wrote 28 original GRE-style verbal questions (fill-in-the-blank, logical reasoning and short reading passages), plus 16 spares in case any turn out unclear in the pilot. Question sets are rotated across rounds so that difficulty is balanced.

3 What we will measure

Each research question from our proposal maps to one comparison and a few numbers the app records on its own.

Research question	What we compare	Main numbers
RQ1: Does a real AI reduce mental effort like the paper's scripted one?	Warm-up (no AI) vs. round 1 with StudyBot, everyone	Effort rating (1–10, same question as the paper), accuracy
RQ2: Does repeated easy help make students lean on the AI?	Standard group, round 1 $\rightarrow$ round 3	How often they try before asking, how often they copy the AI, accuracy when working alone
RQ3: Does the adaptive tutor prevent that without making it harder?	Standard vs. Adaptive group	Accuracy when working alone, wrong answers accepted, effort ratings, NASA-TLX workload

Table 2: Research questions, comparisons and measures.

The main outcome is **accuracy on the solo checks**, because it shows whether students can still reason without the AI. The paper's eye-tracking is replaced by self-rated effort and timing, since we have no eye tracker. For analysis we will use paired tests for RQ1, and mixed-effects models with round (and group, for RQ3) as factors for RQ2 and RQ3.

4 Who, when, and what is left to do

We need about **30 students** (15 per group), each for one 30-minute sitting. With 3–4 laptops running at once, 10–15 people a day is realistic. People who took the GRE in the past year are excluded. Participation is voluntary with on-screen consent, data is stored under participant IDs only, and anyone can stop at any time.

Dates	Step
Oct 23–24	Pilot with 3 people outside the sample; fix unclear questions
Oct 25	Submit Phase 1: the running app and the experimental plan
Oct 26–28	Data collection
Oct 29–31	Analysis and write-up
Nov 1	Submit Phase 2

Table 3: Schedule.

Tasks before October 25

- ☐ Proofread the 28 questions and their correct answers
- ☐ Get an AI API key for the live tutor
- ☐ Run the pilot and note how long it takes
- ☐ Book a room and 3–4 laptops for October 26–28
- ☐ Post the recruitment message in class groups
- ☐ Ask the instructor whether ethics sign-off is needed
- ☐ Record a 2-minute demo of the app for the submission

5 Decisions we want your input on

These are the choices most worth questioning. Please suggest changes to any of them.

1. **One sitting instead of several days.** It keeps the study doable before November 1, but it only shows early signs of over-reliance. Is this acceptable, or should we try 2 sittings on 2 days?
2. **Wrong-answer checks.** Giving a wrong answer on 2 questions is a mild deception, revealed in the debrief. Are we comfortable with it, and does the instructor need to approve it?
3. **Sample size.** 30 students, or can we reach more?
4. **Adaptive rules.** A guiding question at 75% or more correct, a hint at 50–74%. Do these thresholds seem sensible?
5. **Question difficulty.** Are the questions too easy or too hard for our classmates?
6. **Who owns which task** in the list above.